

(SOLUTIONS) EXAM II

CALCULUS I, FALL 2017

Problem 1

As $f(x)$ is a piecewise-defined function, observe that $f(x) = -x^2$ for $x < 0$ and $f(x) = x^2$ for $x \geq 0$. As both $-x^2$ and x^2 are polynomial functions, and as polynomial functions are differentiable everywhere, $f(x)$ is certainly differentiable on $(-\infty, 0)$ and $(0, \infty)$. So, we need only confirm that $f(x)$ is differentiable at $x = 0$ to determine whether $f(x)$ is differentiable everywhere. Using the definition

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$$

we may see that $f(x)$ is differentiable at $x = 0$ if this limit exists. As a two-sided limit exists if both one-sided limits exist and are equal, we then compute

$$\begin{aligned} \lim_{h \rightarrow 0^-} \frac{f(0+h) - f(0)}{h} &= \lim_{h \rightarrow 0^-} \frac{-(0+h)^2 - (0)^2}{h} \\ &= \lim_{h \rightarrow 0^-} \frac{-h^2}{h} \\ &= \lim_{h \rightarrow 0^-} -h = 0. \end{aligned}$$

$$\begin{aligned} \lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} &= \lim_{h \rightarrow 0^+} \frac{(0+h)^2 - (0)^2}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{h^2}{h} \\ &= \lim_{h \rightarrow 0^+} h = 0. \end{aligned}$$

So, as both $\lim_{h \rightarrow 0^-} \frac{f(0+h) - f(0)}{h}$ and $\lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h}$ exist and are equal, we may conclude that

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = 0.$$

Therefore, $f(x)$ is differentiable everywhere.

Problem 2

- (a) $\rightarrow$ (II)
 - (b) $\rightarrow$ (IV)
 - (c) $\rightarrow$ (I)
 - (d) $\rightarrow$ (III)
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Problem 3

(D) is the correct answer.

Problem 4**Part (a)**

$$f'(1) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h}$$

Part (b)

$$\begin{aligned} f'(2) &= \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sqrt{1+h+3} - \sqrt{1+3}}{h} \\ &= \lim_{h \rightarrow 0} \left(\frac{\sqrt{h+4} - \sqrt{4}}{h} \right) \left(\frac{\sqrt{h+4} + \sqrt{4}}{\sqrt{h+4} + \sqrt{4}} \right) \\ &= \lim_{h \rightarrow 0} \frac{(h+4) - (4)}{h(\sqrt{h+4} + \sqrt{4})} \\ &= \lim_{h \rightarrow 0} \frac{h}{h(\sqrt{h+4} + \sqrt{4})} \\ &= \lim_{h \rightarrow 0} \frac{1}{(\sqrt{h+4} + \sqrt{4})} \\ &= \frac{1}{(\sqrt{4} + \sqrt{4})} = \frac{1}{2\sqrt{4}} = \frac{1}{4} \end{aligned}$$

Part (c)

The equation of the line tangent to the graph of $y = f(x)$ at $x = 1$ is given by $y - f(1) = f'(1)(x - 1)$. Computing, this produces

$$y - 2 = \frac{1}{4}(x - 1).$$

Problem 5**Part (a)**

As the positive s (in feet) at time t (in seconds) of the object in question is given by the equation $s(t) = 2t^3 - 3t^2 - 12t + 2$ for $t \geq 0$, the velocity of the object is given by

$$v(t) = s'(t) = 6t^2 - 6t - 12.$$

To find the time t at which the velocity of the object is zero, we solve $v(t) = 0$ as follows:

$$\begin{aligned}v(t) &= 0 \\6t^2 - 6t - 12 &= 0 \\t^2 - t - 2 &= 0 \\(t + 1)(t - 2) &= 0 \\t &= -1 \text{ or } t = 2\end{aligned}$$

As we consider only $t \geq 0$, we then yield that the velocity of the object is zero at time $t = 2$.

Part (b)

The acceleration of the object in question is given by

$$a(t) = v'(t) = s''(t) = 12t - 6.$$

Therefore, by the answer to part (b), the acceleration of the object at the instant the velocity of the object is zero is

$$a(2) = 12(2) - 6 = 18.$$

Part (c)

To find the time at which the acceleration of the object in question is zero, we solve $a(t) = 0$ as follows:

$$\begin{aligned}a(t) &= 0 \\12t - 6 &= 0 \\12t &= 6 \\t &= \frac{1}{2}\end{aligned}$$

Therefore, we conclude that the object has zero acceleration at time $t = \frac{1}{2}$.

Problem 6**Part (a)**

$$f'(x) = \frac{1}{3}x^{-2/3} + 10x - 2$$

Part (b)

$$g'(x) = 2xe^x + (1 + x^2)e^x$$

Part (c)

$$h'(x) = \frac{x\cos(x) - \sin(x)}{x^2}$$

Problem 7**Part (a)**

$$\frac{dy}{dx} = 2(2)\cos(x)(-\sin(x)) = -4\cos(x)\sin(x)$$

Part (b)

$$\frac{dy}{dx} = 3e^{3x}\sec^2(e^{3x})$$

Problem 8

(a)

$$A'(1) = f'(1)g(1) + f(1)g'(1) = 22$$

(b)

$$B'(2) = f'(g(2))g'(2) = -30$$

(c)

$$C'(3) = g'(f(3))f'(3) = -3$$

(d)

$$D'(4) = \frac{f'(4)g(4) - f(4)g'(4)}{(g(4))^2} = -7$$

Problem 9

$$\frac{dy}{dx} = \tan(x)\sec(x)$$

$$\frac{d^2y}{dx^2} = \sec(x)(\tan^2(x) + \sec^2(x))$$

Problem 10**Part (a)**

$$\begin{aligned}\frac{d}{dx}[y^2 + 3xy + x + 6] &= \frac{d}{dx}[0] \\ 2y\frac{dy}{dx} + 3y + 3x\frac{dy}{dx} + 1 &= 0 \\ 2y\frac{dy}{dx} + 3x\frac{dy}{dx} &= -3y - 1 \\ (2y + 3x)\frac{dy}{dx} &= -3y - 1 \\ \frac{dy}{dx} &= \frac{-3y - 1}{2y + 3x}\end{aligned}$$

Part (b)

Substituting $x = 2$ and $y = -2$ into $y^2 + 3xy + x + 6 = 0$ yields

$$(-2)^2 + 3(2)(-2) + (2) + 6 = 0$$

As this is a true statement, the point $(2, -2)$ is on the graph of the curve defined by the given equation.

Part (c)

First, let us note that, at the point $(2, -2)$,

$$\frac{dy}{dx} = \frac{-3(-2) - 1}{2(-2) + 3(2)} = \frac{5}{2} = \frac{5}{2}.$$

The equation of the line tangent to the graph of the relation at the point $(2, -2)$ is given by

$$y + 2 = \frac{5}{2}(x - 2).$$